

Qn	Suggested Answer
Q1	
(a)(i)	The absolute uncertainty of both measurements is the same. OR Fractional/percentage uncertainty is reduced by measuring N coins. By measuring N coins, the absolute uncertainty of the thickness of 1 coin is divided by N. OR By measuring N coins, fractional uncertainty is $\Delta x / T$. By measuring 1 coin, fractional uncertainty is $\Delta x / T/N$.
(a)(ii)	Use a micrometer screw gauge/vernier caliper as the instrument has smaller absolute uncertainty compared to the half metre rule.
(b)	$pV = nRT$ $p = \frac{0.00200 \times 8.31 \times (273.15 + 36.7)}{\frac{4}{3}\pi(0.0250)^3}$ = 78682 Pa $\pm \frac{\Delta p}{p} = \pm \left(3 \frac{\Delta d}{d} + \frac{\Delta T}{T} \right)$ $= \pm \left[3 \left(\frac{0.1}{50.0} \right) + \frac{0.1}{(273.15 + 36.7)} \right]$ $\pm \Delta p = \pm 497 \text{ Pa}$ $p \pm \Delta p = (78700 \pm 500) \text{ Pa}$